

Online Retail Intelligence

Full-Stack Customer and Revenue Analysis for UAE E-Commerce

Business Requirements Document | Version 1.0 |

Prepared By	Role	Document Type	Version
Swathi Anilkumar Sreelatha	Data Analyst	Business Problem Statement	1-6 Phases Completed

1. EXECUTIVE CONTEXT

The global e-commerce sector is growing rapidly, and the UAE sits at its forefront. Ranked among the top 10 fastest-growing e-commerce markets worldwide, the UAE sector is projected to exceed USD 9.2 billion by 2026 (Statista, 2024). UAE consumers demonstrate some of the highest digital spending rates across the MENA region, with a strong preference for gifting, home decor, and lifestyle products, all of which are core categories within this dataset.

In this project, I am simulating the role of a Data Analyst engaged by a UAE-based online retailer. The business has provided one year of transactional data and assigned the analyst to address a set of commercially critical questions:

- Where is revenue being made and where is it being lost?
- Which customers are most valuable, and which are at risk of lapsing?
- At which point in the purchase journey are buyers dropping out?
- How can these findings be converted into concrete, actionable business decisions?

This document sets out the formal business problem framing, analytical questions, dataset overview, methodology, and expected outputs. It serves as the foundation for the full-stack analysis that follows in subsequent phases.

2. BUSINESS PROBLEM STATEMENT

A UAE-based online retailer operating across 38 international markets is generating strong top-line revenue (GBP 10.66M over 13 months), but currently lacks the visibility needed to understand the

customer behaviours, product dynamics, and seasonal patterns that drive, and increasingly threaten, that performance.

The business requires a Data Analyst to identify high-value customer segments, quantify drop-off at each stage of the purchase funnel, forecast seasonal demand, and surface actionable recommendations to inform marketing, inventory, and customer retention strategy.

3. KEY BUSINESS QUESTIONS

This analysis is structured around seven commercially grounded questions, each mapped to a specific analytical technique and a tangible business deliverable.

#	Business Question	Analysis Type	Deliverable
1	Which customers generate the most revenue, and who is at risk of churning?	RFM Segmentation + K-Means Clustering	Customer segments with targeted retention strategy
2	Where are customers dropping off in the purchase journey?	Funnel Analysis	Conversion rates by stage with revenue leakage estimate
3	What drives the Q4 revenue spike, and how can it be maximised?	Time-Series + Seasonal Decomposition	Revenue forecast for next 3 months with stocking recommendations
4	Which products and categories drive revenue, and which underperform?	Product Revenue Analysis	Top and bottom SKU ranking with discontinuation candidates
5	How does customer retention change over time?	Cohort Retention Analysis	Monthly retention heatmap showing loyalty trends
6	What is the revenue opportunity from guest and unregistered customers?	Customer ID Null Analysis	Email capture ROI estimate with guest conversion strategy
7	Which international markets have growth potential beyond the UK?	Geographic Revenue Analysis	Country-level revenue ranking with GCC market opportunity

4. DATASET OVERVIEW

530,103 Transaction Rows	4,338 Unique Customers	38 Countries Covered	GBP 10.66M Total Revenue	13 Months Dec 2010 - Dec 2011	3,684 Unique SKUs
------------------------------------	----------------------------------	--------------------------------	------------------------------------	---	-----------------------------

Source: UCI Machine Learning Repository - Online Retail Dataset (publicly available)

Data Preparation: Cleaned and prepared by Swathi Anilkumar Sreelatha

The original dataset contained 541,909 rows. After removing cancelled orders (Invoice_no prefix 'C'), zero-price entries, and non-product stock codes, the cleaned dataset contains 530,103 rows ready for analysis.

Key Data Quality Finding: 132,219 rows (24.9%) contain no Customer_ID, representing GBP 1.74M in revenue (16.3% of total). These have been classified as guest or unregistered customers. This is a significant business finding in its own right: a substantial volume of revenue cannot be attributed to identifiable customers, presenting a clear email capture and CRM opportunity for the retailer.

5. DATA DICTIONARY

Column	Description	Data Type	Cleaning Applied
Invoice_no	Unique transaction identifier	Integer	Removed cancelled orders (prefix 'C')
Stock_code	Product SKU code	String	Removed non-product codes (POST, AMAZONFEE)
Description	Product name	String	Standardised casing; removed blanks
Quantity	Units purchased per line item	Integer	Removed negative quantities (returns)
Invoice_date	Date and time of transaction	Datetime	Parsed to datetime64; extracted Month and Year
Unit_price	Price per unit in GBP	Float	Removed zero and negative prices
Total_revenue	Quantity x Unit_price	Float	Calculated as a derived column
Customer_ID	Unique customer identifier	Float	24.9% null - flagged as guest customers
Country	Customer country	String	Standardised naming; 38 unique countries
Month_name	Full month name (derived)	String	Extracted from Invoice_date
Year	Transaction year (derived)	Integer	Extracted from Invoice_date
Month_short	3-letter month abbreviation (derived)	String	Extracted from Invoice_date

6. METHODOLOGY AND ANALYTICAL APPROACH

This project follows a structured, end-to-end analytical workflow covering data preparation through to executive-level visualisation. The approach mirrors how a Data Analyst would operate within a UAE commercial team.

Phase 1 - Data Foundation (Excel)

- Raw data ingestion and structural review
- Data cleaning: removed cancellations, returns, zero-price rows, and non-product SKUs
- Feature engineering: derived Total revenue, Month name, Year, and Month short
- Exploratory analysis: pivot tables for revenue by month, country, year, and top products
- Data quality flag: documented 24.9% null Customer as guest customers (GBP 1.74M in revenue)

Phase 2 - Full-Stack Python Analysis (Jupyter Notebook)

- Revenue analysis: monthly trend, country-level contribution, top 10 products by revenue, and a KPI summary table covering total revenue (GBP 10.24M), AOV, unique customers, and segment split
- RFM segmentation: Recency, Frequency, and Monetary scoring per identified customer; Elbow Method applied to determine optimal cluster count (k=4)
- K-Means clustering (sklearn) to segment customers into four tiers: Champions, Loyal Customers, At-Risk, and Hibernating; Champions identified as wholesale and B2B accounts (average spend GBP 125,707) and treated separately from retail segmentation
- Cohort retention analysis: monthly heatmap tracking return rates by first-purchase cohort; December 2010 cohort identified as strongest (approximately 50% retention at Month 12); majority of cohorts show a sharp drop-off after Month 1, confirming one-time buyer risk
- Basket analysis: average items per order calculated by customer segment to quantify purchasing behaviour differences across tiers
- Product affinity analysis: identified top co-purchased product pairs using combinatorial pair counting (itertools); Jumbo Bag and Regency Teacup collections dominate cross-sell opportunities
- Revenue forecasting: 3-month moving average for trend smoothing; Polynomial Regression model (degree=2, sklearn) with Time_Index and Month features; model achieves R-squared of 0.894, forecasting the next 3 months with a +/-5% confidence band
- Guest customer analysis: segmented revenue, order count, and AOV for guest vs. identified customers; guests generate 2.3x higher AOV (GBP 1,099 vs. GBP 475); revenue opportunity model quantified the conversion impact at AED 1.73M
- All analytical outputs exported as structured CSVs for direct Power BI integration with auto-refresh capability

Phase 3 - Funnel Analysis (Python / Plotly)

- Five-stage purchase funnel defined and quantified: All Customers (4,334) > Purchased (4,334) > Repeat Buyers (2,829 | 65.3%) > High Value Top 20% (867) > Loyal Champions RFM Top 20% (649)
- Critical revenue leakage identified: a 69.4% drop-off at the Repeat to High Value transition, representing the largest single point of commercial loss in the customer lifecycle
- Country-level funnel segmentation: Belgium (75%), Germany (72.3%), and France (67.8%) all outperform the UK (65.6%) on repeat conversion rate, signalling higher international retention ROI
- Revenue impact quantified: converting just 10% of 1,505 one-time buyers to repeat purchasers would generate an estimated incremental GBP 367K (AED 1.73M); repeat customers spend 6.9x more than one-time buyers (GBP 2,866 vs. GBP 416)
- Interactive Plotly funnel chart built and exported as funnel.html for direct Power BI embedding; three structured summary DataFrames exported as CSVs for dashboard auto-refresh

Phase 4 - SQL Layer (MySQL 8.0)

- 522,360 rows loaded into MySQL 8.0 via LOAD DATA INFILE with data type validation, NULL handling for guest customers, and revenue field sanitisation
- 12 business queries written covering: monthly revenue MoM growth, top customer ranking, country revenue with cumulative Pareto percentage, repeat rate by country, top products, order frequency distribution, AOV by country, segment performance, and monthly ranking within year
- Window functions applied throughout: LAG () for MoM change, RANK () and DENSE_RANK () for customer and country rankings, SUM () OVER () for cumulative revenue share, NTILE (4) for customer value tier segmentation, and PARTITION BY Year for within-year monthly rankings
- CREATE OR REPLACE VIEW (monthly_revenue_summary) built for reusable reporting; Power BI connected directly via MySQL connector for live dashboard refresh
- SQL-only findings not available in Python: UAE ranked 30th of 38 markets (GBP 5,594 - an untapped GCC opportunity); Netherlands AOV of GBP 9,157 vs. UK GBP 1,310 confirming a wholesale market; Platinum tier spends 36x more than Bronze (GBP 19,142 vs. GBP 529)

Phase 5 - Power BI Executive Dashboard

- Data sourced via MySQL connector (live connection) and three structured CSV files exported from Python; dashboard designed for auto-refresh as new transaction data becomes available
- Validation measures built first to cross-check Power BI totals against Python and SQL outputs, ensuring data integrity across all three analytical layers before building any client-facing visuals
- Advanced DAX measures developed: MoM revenue growth %, Customer Lifetime Value estimate, Repeat Purchase Rate, Revenue per Segment, Guest vs. Identified AOV comparison, funnel stage conversion rates, and AED-converted incremental revenue KPI card
- 3-page interactive dashboard: Executive Summary (headline KPIs, monthly trend, country breakdown), Customer Intelligence (RFM segments, funnel, cohort retention, guest analysis), and Product Performance (top SKUs, affinity pairs, basket analysis)
- Actionable business recommendations embedded directly within each dashboard page, translating analytical findings into commercial decisions for non-technical stakeholders

Phase 6 - Documentation and GitHub Portfolio

- Structured GitHub repository with README, SQL file, notebooks, and .pbix dashboard
- Loom video walkthrough of Power BI dashboard (2 to 3 minutes)
- LinkedIn post targeting UAE Data Analyst market

7. DELIVERABLES TRACKER

Phase	Deliverable	Tools	Status
1 - Data Foundation	Cleaned dataset (530K rows), Data Dictionary, Business Problem Statement	Excel, Python	Complete
2 - Python Analysis	Revenue analysis, RFM K-Means segmentation, Cohort retention heatmap, Product affinity analysis, Polynomial regression forecast (R2=0.894), Guest customer analysis,	Python (pandas, numpy, sklearn, matplotlib, seaborn, Plotly)	Complete

	Basket analysis - all exported as CSVs for Power BI		
3 - Funnel Analysis	5-stage purchase funnel with quantified drop-off rates; revenue leakage of AED 1.73M identified at Repeat to High Value stage; country-level funnel segmentation; interactive Plotly chart exported for Power BI	Python (Plotly)	Complete
4 - SQL Layer	12 business queries in MySQL 8.0 using window functions (LAG, RANK, DENSE_RANK, NTILE, SUM OVER, PARTITION BY); CREATE VIEW for Power BI reporting; UAE market and wholesale tier findings surfaced	MySQL 8.0	Complete
5 - Dashboard	Executive Power BI dashboard - 3 pages: Summary, Customer Intelligence, Products	Power BI, DAX	Complete
6 - Documentation	GitHub repo with README, findings PDF, Loom video walkthrough	GitHub, Markdown	Complete

8. EXPECTED BUSINESS IMPACT

The outputs of this analysis are designed to inform specific, measurable business decisions across five key areas:

Area	Impact and Opportunity
Customer Retention	Identifying at-risk customer segments enables targeted re-engagement campaigns. Even a 5% improvement in repeat purchase rate could add GBP 400K or more in annual revenue, based on current customer-level averages.
Inventory and Supply Chain	The Q4 seasonal spike (September to December accounts for 45% of annual revenue) is a clear signal for pre-season stock planning. Forecasting demand 3 months ahead reduces both stockouts and overstock costs.
Guest Customer Conversion	GBP 1.74M in revenue from unregistered customers presents a high-value CRM opportunity. A targeted email capture strategy at checkout could convert even 10% of these to registered accounts, generating significant long-term value.
Geographic Diversification	The UK currently accounts for 84.6% of revenue, representing a significant concentration risk. The analysis identifies which international markets, including UAE, Australia, and EIRE, have meaningful growth potential to diversify the revenue base.
Product Rationalisation	SKU-level revenue analysis identifies underperforming products suitable for discontinuation, freeing capital and shelf space for high-performing lines.

9. TOOLS AND TECHNOLOGIES

Tool / Technology	Purpose	Skill Level
Microsoft Excel	Data cleaning, pivot tables, EDA visualisations, data dictionary	Advanced
Python (pandas, NumPy)	Data manipulation, feature engineering, statistical analysis	Proficient
Python (matplotlib, seaborn, Plotly)	Visualisation: charts, heatmaps, funnel diagrams	Proficient
Python (sklearn, K-Means)	RFM segmentation and customer clustering	Proficient
Python (sklearn PolynomialFeatures, LinearRegression)	Regression-based revenue forecasting	Proficient
MySQL 8.0	Structured querying, window functions, business queries	Advanced
Power BI + DAX	Executive dashboard, KPI reporting, interactive visuals	Advanced
GitHub	Version control, project documentation, portfolio showcase	Proficient

Document prepared as part of a portfolio project targeting UAE Data Analyst roles. All analysis is conducted on publicly available data. Insights and recommendations are framed as business simulations to demonstrate commercial analytical thinking.